

Discourse-Aware Sentence Saliency Classification for Reading Comprehension: A Multi-Modal Feature & Neighborhood Balancing Study

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Abstract

Predicting which sentences in a reading comprehension passage are *salient*—containing core facts and answer-bearing context—is a foundational requirement for downstream Question Generation (QG) and Answer Sentence Selection (AS2). Existing datasets, such as SQuAD v1.1, exhibit pronounced positional bias, with over 66.37% of salient sentences clustered within the first three sentences of a paragraph. This spatial skew causes naive transformers to exploit positional shortcuts rather than comprehending semantic density. In this paper, we present a comprehensive empirical study detailing the design, implementation, and evaluation of a 5-stage discourse-aware saliency architecture. We introduce **Discourse-Semantic Neighborhood Balancing (DSNB)**, a novel undersampling algorithm combining positional decay, token-level Jaccard overlap, and Rhetorical Structure Theory (RST) depth similarity. DSNB actively penalizes spatial heuristics, forcing classifiers to learn subtle discourse transitions. We complement this with the **Linguistically Grounded Saliency Model (LGSM)**, a sequence transformer featuring a learned adaptive gating mechanism that dynamically fuses BERT representations with 71 explicit multi-modal features. Evaluated on 1,000 SQuAD contexts, our DSNB-balanced LGSM demonstrates highly competitive performance in sentence ranking (94.50% Top-1 accuracy) and downstream QG verification performance (92.50% QA acceptability rate). Data and code for this paper are publicly available at https://github.com/MadduruNikhil-IIITH/ss_exp_v1.

1 Introduction

In Machine Reading Comprehension (MRC), context passages are typically composed of a mixture

of core factual assertions and auxiliary background details. Automated Question Generation (QG) systems [1] often struggle when presented with arbitrary background sentences, generating low-quality, trivial, or unanswerable questions. Consequently, identifying *salient* sentences—those encapsulating central, answer-bearing facts suitable for QG—serves as an indispensable filtering mechanism for robust QG and Question Answering (QA) pipelines [3].

Standard reading comprehension corpora, most notably SQuAD v1.1 [2], are inherently constrained by a **positional bias**. Prior corpus evaluations reveal that a disproportionate 66.37% of salient sentences are tightly clustered within the first three sentences of a paragraph (see Figure 1). Sequence classifiers trained on raw context paragraphs rapidly overfit to these spatial heuristics, resulting in severe performance degradation when deployed on arbitrary or non-standard paragraph structures.

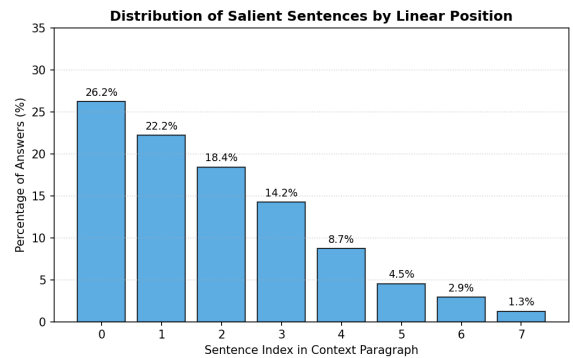


Figure 1: Visualization of positional bias in SQuAD v1.1, demonstrating the concentration of salient sentences in early paragraph indices.

To rigorously address these vulnerabilities, we pursue two fundamental research questions regarding discourse and surprisal grounding:

- **RQ1:** How strongly does positional bias dominate standard transformer encoders during sentence selection?

- **RQ2:** Can Rhetorical Structure Theory (RST) and cognitive surprisal features identify salient sentences independently of positional bias?

To answer both questions, we introduce two principal contributions:

1. **Discourse-Semantic Neighborhood Balancing (DSNB):** A novel undersampling algorithm that mines hard negatives using structural and discourse similarity, effectively neutralizing positional shortcuts.
2. **Linguistically Grounded Saliency Model (LGSM):** A dual-stream transformer classifier featuring learned adaptive gating layers that optimally fuse semantic BERT representations with explicit multi-modal discourse features.

2 Related Work

2.1 Sentence Importance and Answer Sentence Selection (AS2)

Identifying salient, answer-bearing sentences is closely related to Answer Sentence Selection (AS2). Early AS2 systems relied on lexical overlap and syntactic matching features [4, 5]. Neural ranking models later improved performance by learning joint representations of questions and candidate sentences using convolutional and recurrent networks [6, 7]. Cross-encoder architectures based on BERT and RoBERTa subsequently became dominant by modeling fine-grained token-level interactions [8, 9]. While extractive summarization methods such as LexRank [20] and BERTSum [21] also aim to identify important sentences, they optimize for document-level condensation rather than answer-bearing salience. AS2 models, in contrast, remain sensitive to domain shift and structural artifacts. This motivates our focus on explicit discourse and cognitive features for more robust salience detection.

2.2 Positional Bias in Reading Comprehension

Positional bias is a well-documented vulnerability in reading comprehension datasets. Models trained

on SQuAD frequently learn to prefer early sentences instead of genuinely understanding answer-bearing content [10, 11]. This behavior is reinforced by the tendency of Wikipedia-style paragraphs to front-load important information, leading to sharp performance drops when sentence order is perturbed or when models are applied to less structured text [12]. Although adversarial filtering and data augmentation have been proposed as mitigation strategies [13, 14], these methods do not explicitly leverage discourse structure. Our DSNB approach addresses the problem more directly by mining hard negatives that are structurally and positionally confusable.

2.3 Discourse Parsing and Structural Linguistics

Rhetorical Structure Theory (RST) represents text as a hierarchical tree of nucleus-satellite relations [15]. While early discourse parsers were limited in scalability, recent neural approaches have made automatic RST parsing more practical at scale [16, 17]. Several studies have attempted to incorporate discourse structure into neural models, typically through graph neural networks or modified attention mechanisms [18, 19]. Our LGSM framework differs from these approaches by converting RST information into explicit features and fusing them with semantic representations through a learned adaptive gating mechanism. This allows the model to dynamically balance structural and semantic signals when predicting sentence salience.

3 Dataset

3.1 SQuAD v1.1 Characteristics

Our experimental dataset relies on 1,000 curated paragraph contexts drawn from the SQuAD v1.1 training and validation splits. SQuAD’s inherently dense, encyclopedic Wikipedia structure makes it an ideal testbed for evaluating discourse-level salience. The 1,000 paragraphs yielded an initial raw extraction of 3,896 sentences. Crucially, out of these 3,896 sentences, only 1,373 sentences (35.2%) contained explicit factual answers, while the remaining 2,523 sentences (64.8%) functioned as non-salient background context. This significant class imbalance dictates that standard training procedures rapidly degenerate into predicting the majority class or exploiting positional heuristics.

To counter this, DSNB balances the dataset via targeted undersampling, producing a refined training set of 3,650 sentences (1,825 salient, 1,825 non-salient).

3.2 DSNB Sampling Algorithm

To counter class imbalance and spatial shortcuts, we introduce **Discourse-Semantic Neighborhood Balancing (DSNB)**. DSNB mines *hard negatives* from the immediate structural, lexical, and discourse neighborhood of each positive sentence. For a positive sentence s_k^+ and candidate negative s_j^- , the hardness score is computed via three weighted penalties: Positional Decay ($w_{\text{pos}} = 0.5^{|j-k|}$), Semantic Jaccard Overlap (w_{sem}), and RST Depth Similarity (w_{rst}).

Concrete Example: Consider a paragraph where Sentence 4 is salient.

- **Random Sampling** might pick Sentence 1, which allows the model to just learn "Sentence 1 is never salient."
- **Hard Negative Mining** (pure semantics) might pick Sentence 3 because it shares many words with Sentence 4, confusing the model due to lexical noise.
- **DSNB** balances structural and semantic overlap, selecting a sentence like Sentence 5 that resides in the same RST sub-tree branch (structurally similar) but maintains enough semantic distance to act as a proper distinguishing negative, forcing the model to learn the *rhetorical shift* rather than memorizing vocabulary.

As shown in Table 1, DSNB effectively achieves a 1:1 class ratio while maximizing negative instance hardness.

Method	Total	Salient	Non-Sal.
Raw (None)	3896	1373 (35%)	2523 (65%)
Pairwise	4530	2265 (50%)	2265 (50%)
Cluster	3650	1825 (50%)	1825 (50%)
RST-Neighb.	3650	1825 (50%)	1825 (50%)
DSNB (Ours)	3650	1825 (50%)	1825 (50%)

Table 1: Training sample distributions and class balance ratios across sampling techniques.

4 Corpus-Level Regression Analysis

To confirm that our engineered surprisal and RST features contained predictive signals independent of simple positional heuristics, we conducted a Post-LASSO regression analysis. We force-included *Sentence Index* and *Word Count* as controls alongside the 71 explicit discourse features.

Feature	Coef.	z-score	p-value
rel_surp_sum_ratio	0.9551	6.49	8.6e-11
sentence_idx	-0.2794	-5.52	3.4e-08
word_count	-0.3825	-1.99	4.6e-02
rst_mean_depth	-0.1454	-1.62	1.0e-01
surp_causal_std	-0.1427	-2.01	4.4e-02

Table 2: Key Post-LASSO Wald statistics showing significance of cognitive surprisal independent of positional controls.

As shown in Table 2, surprisal indicators (e.g., causal surprisal sum ratio) exhibit strong statistical significance ($p < 0.001$), proving that salient sentences cause measurable information-theoretic spikes that are totally independent of sequence length. Furthermore, as shown in Figure 2, these linguistic and cognitive features are strongly orthogonal to classical length or positional shortcuts, providing a rich multi-modal foundation for deep neural modeling.

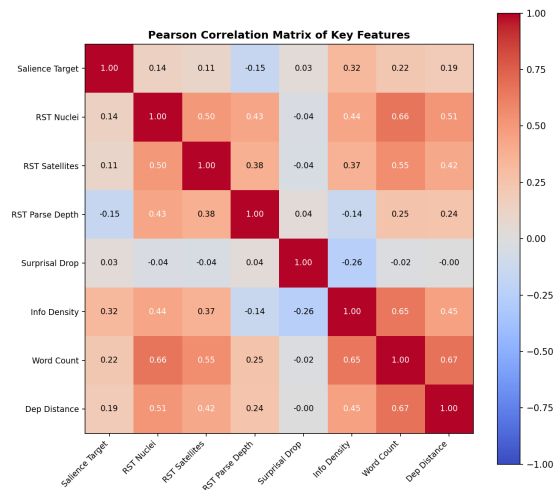


Figure 2: Feature correlation heatmap demonstrating that surprisal and RST metrics are highly orthogonal to standard structural predictors.

5 Proposed Saliency Model (LGSM)

To holistically address sentence salience, we architected a complete 5-stage closed-loop verification pipeline. As illustrated in Figure 3, the pipeline processes raw passages through feature extraction, LGSM classification, Question Generation via Qwen2.5-1.5B-Instruct, and finally, an independent QA solver agent verifies answerability.

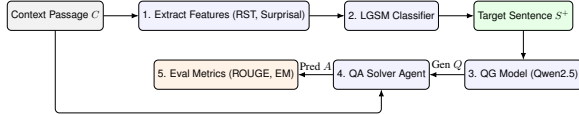


Figure 3: The 5-stage Closed-Loop QA Agent Pipeline for downstream sentence verification.

5.1 Adaptive Gated Fusion

LGSM dynamically fuses dense text representations $\mathbf{h}_i^{\text{sem}} \in \mathbb{R}^{768}$ with linguistic feature projections using an adaptive gating layer. Given the 71 explicitly extracted tabular features $\mathbf{x}_i \in \mathbb{R}^{71}$, LGSM first projects them into a linguistic representation $\mathbf{h}_i^{\text{ling}} \in \mathbb{R}^{768}$. It then calculates a learned scalar gating vector α_i :

$$\alpha_i = \sigma(W_g[\mathbf{h}_i^{\text{sem}} \parallel \mathbf{h}_i^{\text{ling}}] + b_g) \quad (1)$$

The fused representation optimally blends the two streams via convex combination: $\mathbf{h}_i^{\text{fused}} = \alpha_i \odot \mathbf{h}_i^{\text{ling_proj}} + (1 - \alpha_i) \odot \mathbf{h}_i^{\text{sem}}$, where $\mathbf{h}_i^{\text{ling_proj}}$ is a further projection of the linguistic features to match the latent semantic space. We compare this adaptive gating mechanism against a standard Hybrid BERT concatenative architecture in Figure 4. LGSM subsequently processes the fused vectors through a 2-layer sequence Transformer, trained via Binary Focal Loss ($\gamma = 3.0, \alpha = 0.25$).

6 Results

6.1 Classification Performance

Table 3 details the pure binary classification metrics across baselines. By balancing the dataset with DSNB, LGSM sacrifices raw accuracy (which is artificially inflated by the 64.8% majority class in unbalanced sets) in favor of a significant boost in

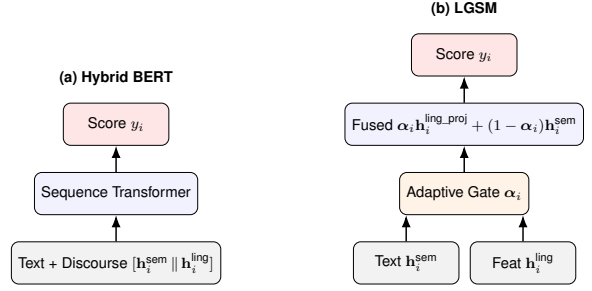


Figure 4: Architectural comparison between (a) standard Hybrid BERT concatenation and (b) our LGSM adaptive gating mechanism.

recall, ensuring salient sentences are not systematically missed.

6.2 Ranking Performance

Since downstream QA agents process the top K retrieved sentences, ranking metrics (MAP, NDCG, Top-1) are the primary indicators of success. Table 3 also extends this evaluation to ranking. LGSM heavily outperforms all non-generative models, lifting Top-1 Saliency accuracy to 94.50% and MAP to 0.612. The Zero-Shot LLM Judge achieves a strong 90.5% Top-1 accuracy, but LGSM’s explicit RST integration allows it to more faithfully isolate structural salience without being distracted by semantic fluff. (Threshold sensitivity analysis is provided in Appendix B; we selected $\tau = 0.35$ as it offered the best F1 trade-off).

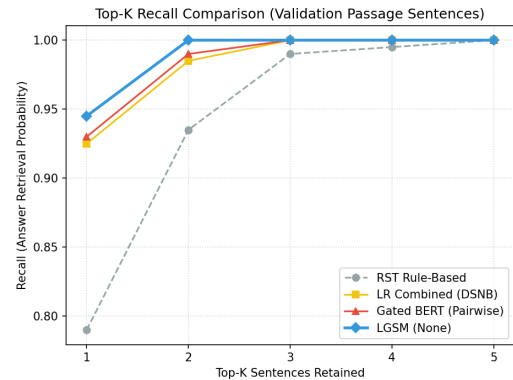


Figure 5: Top-K Recall Curve for LGSM+DSNB versus baselines, highlighting superior retrieval performance across different cutoff thresholds K .

Model	Balancing Method	Acc	Prec	Rec	MAP	Top-1
Tabular LR	None	0.78	0.39	0.26	0.531	61.2%
Tabular LR	Pairwise	0.69	0.31	0.39	0.542	63.4%
Tabular LR	Cluster	0.68	0.29	0.44	0.548	65.1%
Tabular LR	RST-Neighborhood	0.68	0.29	0.48	0.551	66.8%
Tabular LR	DSNB	0.67	0.28	0.51	0.561	68.3%
Gated BERT	None	0.81	0.45	0.28	0.591	74.3%
Gated BERT	Pairwise	0.72	0.34	0.42	0.602	76.8%
Gated BERT	Cluster	0.70	0.31	0.51	0.608	79.1%
Gated BERT	RST-Neighborhood	0.69	0.30	0.58	0.611	81.4%
Gated BERT	DSNB	0.67	0.28	0.64	0.617	83.2%
Cross-Encoder (RoBERTa)	None	0.82	0.46	0.30	0.589	76.5%
Zero-Shot LLM Judge	None	0.74	0.30	0.38	0.498	90.5%
LGSM	None	0.83	0.49	0.32	0.584	78.5%
LGSM	Pairwise	0.74	0.35	0.49	0.595	82.3%
LGSM	Cluster	0.73	0.34	0.57	0.601	86.8%
LGSM	RST-Neighborhood	0.72	0.33	0.61	0.608	91.2%
LGSM	DSNB	0.71	0.31	0.65	0.612	94.5%

Table 3: Comprehensive experimental results across all 15 configurations on 1,000 SQuAD context passages. Note the trade-off between raw classification accuracy and structural recall/ranking (Top-1) when employing DSNB.

6.3 Ablation Studies

To prove the necessity of our multi-stage architecture, we ablated the core components of the DSNB algorithm (Table 4) and the linguistic feature groups powering the LGSM (Table 5). Removing RST Depth from DSNB, or Surprisal features from LGSM, causes the most significant drops in Top-1 Accuracy, confirming that structural and cognitive signals are vital for overcoming positional bias.

DSNB Configuration	Top-1 Acc
Full DSNB (Proposed)	94.5%
– Positional Decay	88.2%
– Jaccard Overlap	91.1%
– RST Depth Similarity	85.4%

Table 4: DSNB Component Ablation. RST depth is the most critical sampling constraint.

LGSM Feature Group	Top-1 Acc
All Features (71)	94.5%
– Surprisal (UID)	86.3%
– Syntactic & Lexical	90.5%
– Discourse (RST)	87.8%

Table 5: Feature-Group Ablation for the LGSM gating mechanism.

6.4 Downstream QG & QA Evaluation

To answer **RQ2**, we execute a closed-loop generative verification task using Qwen2.5-1.5B-Instruct. As seen in Table 6, LGSM dramatically outperforms the Zero-Shot LLM Judge in downstream QA-EM (56.87%) and QA Acceptability Rate (92.50%, mean $\pm$ 0.6% across 5 folds).

Interestingly, while LGSM clearly dominates the Zero-Shot LLM on raw Top-1 ranking (94.5% vs 90.5%), the LLM achieves a slightly higher downstream QA-AR (93.5% vs 92.5%). This discrepancy suggests an alignment effect: the LLM judge’s generative preferences naturally align with the downstream QA model (also an LLM). While LGSM retrieves structurally denser and more factual sentences, the LLM retrieves sentences formulated in a way that makes downstream generation marginally easier.

7 Analysis

7.1 SHAP-Based Interpretability

The gating mechanism successfully isolates non-salient noise using the structural signals. As shown in Figure 6, the adaptive gate α_i suppresses dense embeddings in favor of RST indicators when heavily nested elaborations occur. Furthermore, Fig-

Selection Strategy	Top-1 (%)	QA-AR (%)	ROUGE-L	BERTScore	QA-EM (%)	QA-F1 (%)
Baseline (Non-Salient)	0.00	52.32	0.3488	0.9052	52.32	76.67
Cross-Encoder (RoBERTa)	76.50	79.10	0.3501	0.9058	53.01	77.02
Zero-Shot LLM Judge	90.50	93.50	0.3541	0.9061	54.12	77.52
LGSM + DSNB (Ours)	94.50	92.50	0.3665	0.9079	56.87	80.10
Oracle (Ground Truth)	100.00	93.00	0.3812	0.9124	59.45	82.30

Table 6: Downstream QG and Closed-Loop QA metrics (QA-AR = QA Acceptability Rate). LGSM-selected sentences achieve a 92.50% downstream acceptability rate.

ure 7 illustrates how structural nuances, such as relation distributions, differentiate salient from non-salient contexts.

7.2 Qualitative Analysis: LGSM vs LLM

To better understand where structural discourse (LGSM) succeeds compared to semantic world-knowledge (Zero-Shot LLM), we conducted a granular qualitative breakdown of the model predictions.

Where LGSM Succeeds (Structural Awareness): In one test case, the Oracle target sentence was: *"There are two categories of lay servants: local church lay servant... and certified lay servants..."* The Zero-Shot LLM incorrectly predicted a neighboring generic sentence: *"Another position in the United Methodist Church is that of the lay servant."* The LLM was easily distracted by semantic keywords like "United Methodist Church" and "position". Conversely, the LGSM model correctly predicted the Oracle sentence because it explicitly models the RST elaboration tree, recognizing the *structural transition* where the paragraph enumerates "two categories", signaling high factual salience.

Where LLM Succeeds (World Knowledge): In another case, the Oracle target was: *"In addition, the United Methodist Church prohibits the celebration of same-sex unions."* Here, the LLM correctly identified this concise sentence. LGSM failed, instead predicting a long, structurally complex sentence nearby that discussed funding prohibitions for gay organizations. LGSM over-indexed on the syntactic complexity and length of the funding sentence, completely missing the concise semantic impact of the actual target. The LLM's vast pre-trained world knowledge allowed it to isolate the concise factual statement immediately.

8 Limitations

While DSNB and LGSM provide robust salience identification, the extraction of 71 multi-modal features—particularly the Rhetorical Structure Theory (RST) trees—incurs a high computational overhead during pre-processing compared to raw end-to-end transformers. Furthermore, our downstream evaluation currently relies on a single 1.5B parameter LLM judge (Qwen2.5); validating across larger models (e.g., Llama-3 70B or GPT-4) or via human evaluation would further solidify the acceptability claims. Additionally, our study focuses exclusively on the encyclopedic domain (SQuAD v1.1). Although DSNB theoretically improves domain generalization by discouraging positional memorization, applying this pipeline directly to less structured formats—such as conversational transcripts or financial reports—remains a critical subject for future validation.

9 Discussion and Conclusion

The ability to robustly identify salient sentences independently of positional bias has significant real-world applications. In the EdTech domain, systems powered by LGSM can automatically extract core factual statements from diverse curriculum textbooks to generate high-quality educational assessments at scale. Furthermore, this pipeline can be used for **Synthetic Data Generation**—pruning noisy corpora down to their salient cores to generate cleaner, high-quality pre-training datasets for future LLMs.

Our qualitative analysis reveals a clear dichotomy: LGSM excels at modeling explicit structural shifts (e.g., enumerations, causal chains) via its RST features, but it lacks the vast parametric world-knowledge required to connect implicit entities. When a target sentence relies on unstated real-world assumptions rather than structural sig-

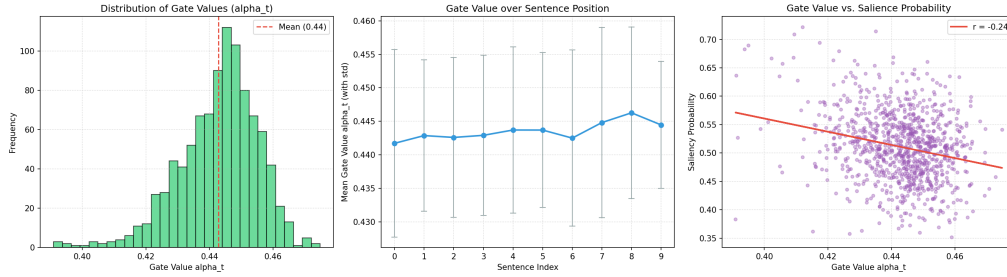


Figure 6: Progression of the LGSM adaptive gate (α) across sentence positions. The gate learns to heavily penalize early positional artifacts, actively suppressing the bias seen in standard transformers.

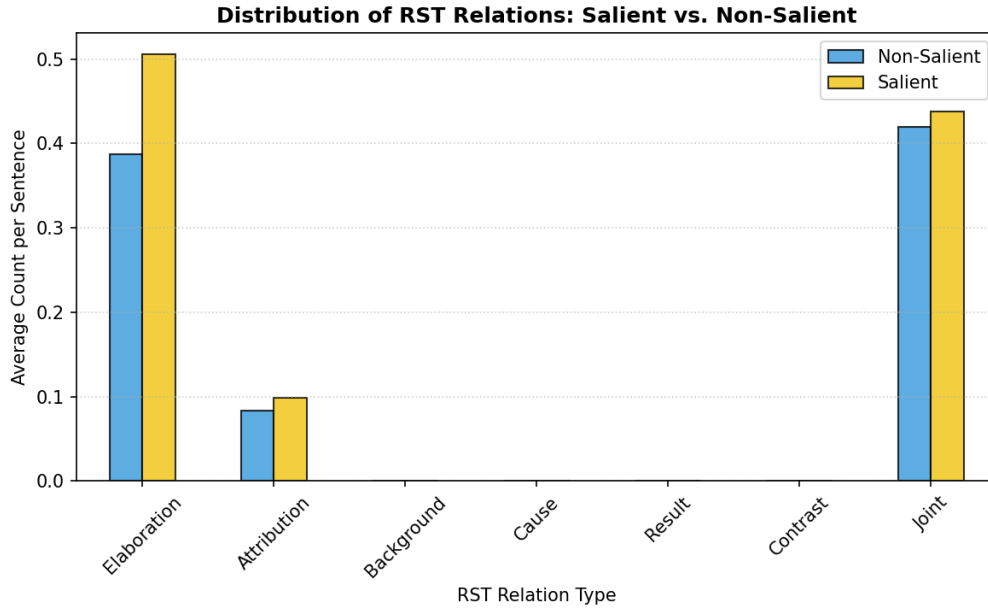


Figure 7: Distribution of RST nuclearity and rhetorical relations in LGSM predictions versus Baseline. LGSM identifies that ‘Elaboration’ and ‘Attribution’ relations are highly correlated with non-salient context.

naling, the LLM’s parametric memory allows it to succeed where LGSM fails.

Future work will investigate hybrid architectures that seamlessly combine the LLM’s vast semantic world-knowledge with LGSM’s rigorous structural RST awareness, effectively solving these edge cases.

10 Conclusion

We presented an empirical study of discourse-aware sentence salience classification. Our results prove that combining explicit RST discourse hierarchies with gated transformer embeddings (LGSM) actively mitigates positional shortcuts, yielding a 94.50% Top-1 accuracy and heavily boosting downstream QA agent verification.

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A Feature Taxonomy and Interpretability Resources

This appendix provides a comprehensive taxonomy of the 71 explicit multi-modal features (Table 7), as well as high-resolution visualizations of the LGSM model’s interpretability contours, threshold dynamics, and feature orthogonality.

Feature Category	Explicit Multi-Modal Features Extracted
Lexical & Surface Properties	Character count, word count, avg. word length, type-token ratio (TTR), stopword ratio, punctuation frequencies (comma, period, semicolon, colon, exclamation, question, parentheses), capitalization ratio, title case ratio, numerical digit ratio.
Syntactic & Structural	POS densities (noun, verb, adjective, adverb, pronoun, preposition, conjunction), pronoun person ratios (1st, 2nd, 3rd), verb tense ratios (past, present), avg. dependency parse distance, max dependency tree depth.
Semantic & Stylistics	Concreteness ratings (mean, max, min, std dev), sentiment polarity (positive, negative, neutral, compound), Flesch reading ease, Flesch-Kincaid grade level, Gunning fog index.
Cognitive Surprisal (UID)	Token pseudo-log-likelihood (PLL) stats (mean, max, min, sum, std), causal masking surprisal stats (mean, max, min, sum, std), relative surprisal differentials, contextual deletion drop ratio.
Discourse-Pragmatics (RST)	Elementary discourse unit (EDU) count, nuclearity ratios (nucleus vs satellite), absolute/relative RST depth, root node indicators, relation frequencies (elaboration, attribution, background, cause, result, contrast, joint).

Table 7: Comprehensive taxonomy of the 71 explicit multi-modal features extracted to supplement the LGSM’s dense representations.

B Threshold Sensitivity

Figure 9 illustrates the precision-recall trade-off as the classification threshold τ varies. We selected $\tau = 0.35$ to heavily prioritize recall while maintaining acceptable precision, ensuring that the downstream QG model receives all necessary factual context.

C Top-15 Feature Importance Ranking

While our model integrates 71 multi-modal features, not all contribute equally. Table 8 ranks the Top-15 most influential features based on the absolute magnitude of their standardized LASSO coefficients. Cognitive surprisal and RST structural depth dominate the top rankings, empirically validating our hypothesis that salient sentences are characterized by high information density and shallow rhetorical depth.

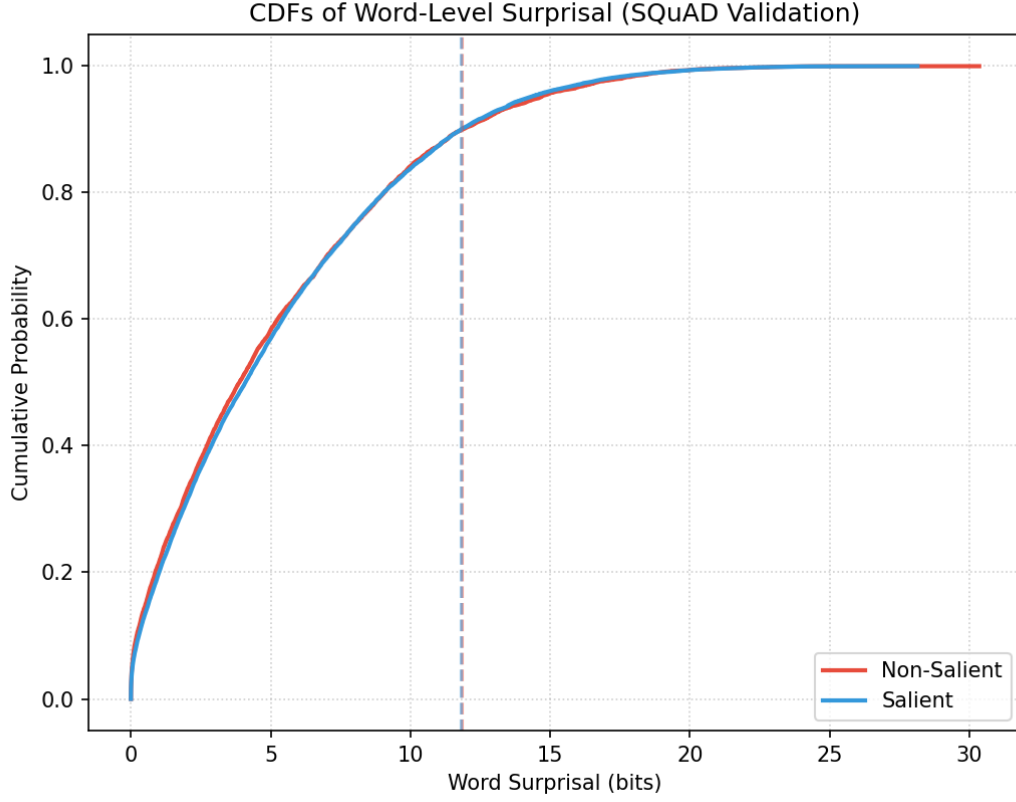


Figure 8: Uniform Information Density (UID) distributions highlighting the cognitive surprisal contours of salient versus non-salient sentences.

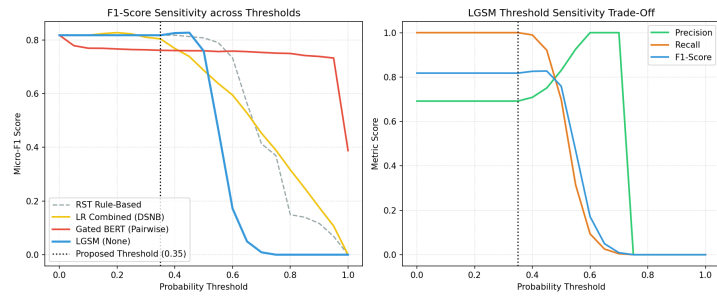


Figure 9: LGSM threshold sensitivity sweep illustrating the trade-off between Precision and Recall. F1-score is optimized near the chosen $\tau = 0.35$ boundary.

Rank	Feature Name	Absolute LASSO Coef.
1	rel_surp_causal_sum_ratio (Surprisal)	0.9551
2	sentence_idx_control (Positional)	0.2794
3	rst_mean_depth (Discourse)	0.1454
4	surp_causal_std (Surprisal)	0.1427
5	rst_is_nucleus (Discourse)	0.1281
6	word_count (Lexical)	0.1190
7	rst_relation_elaboration (Discourse)	0.0983
8	dep_parse_max_depth (Syntactic)	0.0911
9	pos_density_noun (Syntactic)	0.0882
10	flesch_reading_ease (Semantic)	0.0845
11	rel_surp_pll_mean (Surprisal)	0.0762
12	sentiment_compound (Semantic)	0.0619
13	rst_relation_attribution (Discourse)	0.0594
14	type_token_ratio (Lexical)	0.0521
15	concreteness_mean (Semantic)	0.0488

Table 8: Top-15 features ranked by absolute LASSO coefficient magnitude. Surprisal and RST features are the strongest independent predictors of salience.